

Excel Functions Cheat Sheet for Data Analysts & BI Professionals

1. Math & Statistical Functions

Function	Description
SUM(range)	Adds up values in a range
AVERAGE(range)	Returns the mean of a range
MEDIAN(range)	Returns the median
MODE.SNGL(range)	Returns the most frequent value
COUNT(range)	Counts numeric values
COUNTA(range)	Counts non-empty cells
COUNTBLANK(range)	Counts empty cells
ROUND(number, num_digits)	Rounds to specified digits
ROUNDUP() / ROUNDDOWN()	Rounds number up/down
RAND() / RANDBETWEEN(min, max)	Random number(s)

2. Lookup & Reference Functions

Function	Description
VLOOKUP(lookup_value, table, col_index, [range_lookup])	Vertical lookup (older but still common)
HLOOKUP(...)	Horizontal version of VLOOKUP
INDEX(array, row_num, [column_num])	Retrieves a value at a specific position
MATCH(lookup_value, lookup_array, [match_type])	Returns position of value
XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode])	Modern replacement for VLOOKUP/HLOOKUP
OFFSET(reference, rows, cols)	Returns a range offset from a cell
INDIRECT(ref_text)	Returns a reference specified by text
CHOOSE(index, option1, option2, ...)	Picks from a list of options

3. Text Functions

Function	Description
LEFT(text, num_chars) / RIGHT()	Extract characters from left/right
MID(text, start_num, num_chars)	Extract substring from middle
LEN(text)	Counts characters in text
FIND(find_text, within_text)	Finds position of substring (case-sensitive)
SEARCH(find_text, within_text)	Like FIND but not case-sensitive
TEXT(value, format_text)	Format number/dates as text
CONCAT(text1, text2, ...)	Combines text (Excel 2019+)
TEXTJOIN(delimiter, ignore_empty, text1, ...)	Joins multiple text pieces
TRIM(text)	Removes extra spaces
UPPER(), LOWER(), PROPER()	Changes text case
SUBSTITUTE(text, old, new)	Replace all occurrences of text
REPLACE(text, start, num_chars, new_text)	Replace part of a string

4. Logical Functions

Function	Description
IF(logical_test, value_if_true, value_if_false)	Basic conditional logic
IFS(condition1, result1, condition2, result2, ...)	Multiple IFs without nesting
AND(condition1, condition2)	Returns TRUE if all are TRUE
OR(condition1, condition2)	Returns TRUE if any are TRUE
NOT(logical)	Reverses logical value

5. Date & Time Functions

Function	Description
TODAY()	Returns current date
NOW()	Returns current date and time
DATE(year, month, day)	Builds a date
DATEDIF(start_date, end_date, unit)	Returns difference between dates
YEAR(date), MONTH(), DAY()	Extracts date parts
TEXT(date, "mmm")	Formats date as month name
EDATE(start_date, months)	Add/subtract months from date
EOMONTH(start_date, months)	Returns last day of month

6. Data Analysis Functions

Function	Description
UNIQUE(range)	Returns unique values from a list (Excel 365+)
SORT(range, [sort_index], [order])	Sorts a range
FILTER(array, include, [if_empty])	Filters rows dynamically
SUBTOTAL(function_num, range)	Conditional aggregate (ignores hidden rows)
AGGREGATE(function_num, options, range)	Similar to SUBTOTAL with more control
RANK.EQ(number, ref)	Ranks number in dataset
PERCENTILE.EXC(range, k)	Finds k-th percentile
STDEV.P(range) / STDEV.S(range)	Standard deviation (population/sample)

7. Financial & Business Functions (Useful Additions)

Function	Description
NPV(rate, value1, ...)	Net present value of cash flows
IRR(values)	Internal rate of return
PMT(rate, nper, pv)	Loan payment per period
FV(rate, nper, pmt, [pv], [type])	Future value
IFERROR(value, value_if_error)	Handles errors gracefully
ISERROR(value)	Checks for error
CELL(info_type, [reference])	Info about a cell
INFO(type_text)	Info about the environment (e.g., OS)



Bonus Tips

- Use Tables (Ctrl + T) to auto-expand formulas and make ranges dynamic.
- Named Ranges: Use for readability and easier maintenance.
- Dynamic Arrays: Newer Excel versions (365+) let you use formulas that return multiple results automatically (e.g., UNIQUE, FILTER, SORT).
- Power Query: Use for complex transformations, merging, and cleaning without formulas.
- PivotTables: For fast summaries and ad hoc reports.